Industrial Security (Bachelor)

Exercise Sheet – Section 01

Introduction to Industrial Security

Goal of this sheet

Practise placing systems in the IT/OT spectrum, articulating why industrial security is a distinct discipline, and exploring one historical incident in depth.

Exercise 1.1 – Identify the System

For each scenario, decide whether the system is primarily **IT**, **OT**, or a **hybrid (IIoT)**. Justify your answer in 2–3 sentences each.

1. A SQL database storing customer billing records.
2. A wind-turbine pitch controller that streams telemetry over MQTT to a cloud broker.
3. A traffic-light controller at a busy intersection.
4. A workstation running engineering software used to program PLCs.
5. A vibration sensor on a pump that publishes to a corporate Kafka topic.

Exercise 1.2 – CIA Re-ordered

The OT priority order is often written as $A \succ I \succ C$ rather than $C \succ I \succ A$.

1. Give a concrete plant scenario where putting confidentiality first would harm safety.
2. Give a scenario where strong confidentiality is nonetheless important in OT (think trade secrets, recipes, formulations).
3. Sketch a short policy statement (3–5 lines) that an asset owner could put in writing to make the priority explicit to all parties.

Exercise 1.3 – Group Discussion

In groups of four:

1. Pick one critical-infrastructure sector (electricity, water, transport, food, healthcare).
2. List five physical processes within it that depend on digital control.
3. For each process, name the worst-case outcome of an availability incident.
4. Prepare a 3-minute pitch summarising your top concern.

Hint

Useful starting points: pumps and SCADA in water utilities, breakers and protection relays in substations, signalling in rail, sterilisation cycles in pharma.

Exercise 1.4 – Research Mini-Project

Pick one historical incident and write a 1-page summary (about 400 words):

- Maroochy Shire sewage (2000)
- Stuxnet (2010)
- Ukraine power grid – BlackEnergy/Industroyer (2015 & 2016)
- TRITON / TRISIS (2017)
- Norsk Hydro ransomware (2019)
- Colonial Pipeline (2021)
- Oldsmar water treatment (2021)
- CyberAv3ngers / Unitronics (2023)

Your summary must cover: *attacker, target, initial vector, technical steps, physical impact, defensive lessons.* Cite at least two sources (CISA advisory, vendor post-mortem, peer-reviewed paper).

Caution

Do not interact with or scan any real infrastructure. Use only published material.

Exercise 1.5 – Cost of an Incident

For one of the four cost case studies in the slides (Colonial Pipeline, Norsk Hydro, Maersk/NotPetya, Stuxnet), find two public sources and reconstruct:

1. The headline financial figure – what is included (ransom, recovery, business interruption)?
2. One indirect cost often missed (regulatory fines, customer churn, insurance premium increase).
3. One non-financial cost (reputation, employee morale, safety culture).

Exercise 1.6 – Industry 4.0 Risk Tour

For your institution (or a fictional plant) list three Industry 4.0 / IIoT integrations on the roadmap (cloud analytics, digital twin, edge AI, mobile workforce). For each:

- Identify one new attack vector it introduces.
- Propose one compensating control.